

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer Examination – 2024

Course: B. Tech.

Branch: Civil Engineering

Semester: IV

Subject Code & Name: BTCVC405 Hydraulics - II

Max Marks: 60

Date: 24/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain open channel flow and state difference between open channel flow and pipe flow.	Understand	6
B) Derive an expression for discharge through triangular notch with neat diagram.	CO1	6
C) A reservoir $4.65 \times 10^4 \text{ m}^2$ in area is to be controlled by a rectangular weir with its crest level at El. 30 m. it is intended to provide such a length of weir that will lower the water level from El. 31.2 m to El. 30.6 m in half an hour time. Determine the length of the weir. The discharge over the weir is given by the formula $Q = 1.9 LH^{3/2}$, where Q is discharge in cumec, L is crest length in meters and H is head over the weir in meters.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Define the term most economical section of channel. Derive the conditions for the most economical triangular channel section?	CO1	6
B) Define specific energy and give detailed explanation about specific energy curve with neat labelled sketch.	Remember	6
C) Design the most economical trapezoidal canal section having one side vertical and other side slope 1.5 H:1 V to carry the discharge of $10 \text{ m}^3/\text{s}$. Take Manning's coefficient $n = 0.015$ and bed slope $= 1/3500$.	Create	6
Q. 3 Solve Any Two of the following.		12
A) Derive the dynamic equation for gradually varied flow with assumptions in	CO2	6

the form:
$$\frac{dy}{dx} = \frac{S_0 - S_f}{1 - F_r^2}$$

- B)** With usual notation in hydraulic jump, prove the following relation for hydraulic jump in rectangular channel **CO2** **6**

$$\frac{y_2}{y_1} = \frac{1}{2} \left[-1 + \sqrt{1 + 8F_{r1}^2} \right]$$

- C)** A jet of water 5 cm in diameter having a velocity of 25 m/s strikes normally a smooth flat plate. Determine the thrust when **CO3** **6**
- i. The plate is stationary.
 - ii. The plate is moving with a velocity of 5 m/s away from jet along the line of jet.
- What is the work done per second by the jet in each case?

Q.4 Solve Any Two of the following. **12**

- A)** Enlist types of turbines. Describe components of Francis Turbine with neat labelled sketch. **Remember** **6**
- B)** Discuss the criteria considered for the selection of turbines. **Understand** **6**
- C)** Give description about draft tube and its function. **Remember** **6**

Q. 5 Solve Any Two of the following. **12**

- A)** What are different component parts of a centrifugal pump? Explain their function with neat sketch. **Remember** **6**
- B)** Why priming is necessary for centrifugal pump? Enlist & describe priming devices used for priming. **Understand** **6**
- C)** Explain **Remember** **6**
- i) Submersible pump
 - ii) Multistage pump

***** End *****